

Scaling PDF forensics through augmentation and robustness analysis

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Summary

This project extends the ForensicsDetective framework by evaluating whether machine learning classifiers can reliably identify PDF documents under realistic distortions. The original premise of the ForensicsDetective project is that PDF files contain generation-specific binary signatures that can be converted into grayscale images and classified according to their source. Prior baseline experiments showed that these binary-image representations are highly discriminative, with classification accuracy often approaching 100% on clean data. The goal of this assignment was to determine whether those strong results persist after the data was scaled through augmentation and after additional classifiers are introduced into the pipeline.

To accomplish this, five independent augmentations were applied to each original PDF-derived image: Gaussian noise, JPEG compression, DPI downsampling, random cropping, and bit-depth reduction. These augmentations were chosen because they approximate realistic distortions such as recompression, scanning artifacts, reduced quality, and partial document visibility. The augmentation process expanded the dataset to six times its original size, creating a much more challenging evaluation environment while keeping the original training data unchanged. This design made it possible to isolate robustness from simple memorization.

Four classifiers were ultimately evaluated: SVM, SGD, Random Forest, and KNN. Results showed that the best-performing models on clean data were SGD and Random Forest, while KNN performed poorly and behaved close to random guessing. Robustness testing revealed that structural distortions, especially random cropping and DPI downsampling, caused the largest performance degradation. In contrast, Gaussian noise, JPEG compression, and bit-depth reduction had minimal effect on the strongest models. These findings suggest that provenance classification depends more heavily on preserving structural and spatial layout information than on preserving exact pixel intensities.

The strongest overall conclusion is that PDF provenance detection through binary-image representations is highly effective, but robustness depends strongly upon the type of distortion applied. Linear and ensemble-based methods performed best, while distance-based methods struggled in the high-dimensional feature space.

1. Introduction

Digital documents are often treated as visually identical when they contain the same text, but their underlying binary structure can differ dramatically depending on how they were created. A PDF produced by Microsoft Word, for example, may contain distinct internal formatting, object ordering, metadata patterns, and binary organization compared with a PDF exported from Google Docs or generated programmatically in Python. Those differences are usually invisible to human readers, yet they can still leave detectable signatures that may be useful in digital forensics.

The ForensicsDetective project is built on the idea that PDF binary data can be transformed into grayscale images and analyzed as if they were visual patterns. Instead of interpreting the semantic content of the document, the classifier learns the visual signatures of the binary encoding itself. This creates an unusual but powerful machine learning formulation of document provenance detection. The baseline project demonstrated that SVM and SGD classifiers can distinguish document origins with very high accuracy under clean conditions.

However, very high clean-data accuracy does not automatically imply robustness. In real-world conditions, documents may be rescanned, recompressed, cropped, resized, degraded, or otherwise altered before forensic analysis takes place. A useful forensic system must therefore perform well not only on pristine samples but also under controlled distortions that simulate realistic acquisition conditions. This assignment explicitly focuses on that gap by requiring augmentation, robustness testing, additional classifiers, and a full comparative analysis.

The central research question of this work is:

How robust are machine learning classifiers for PDF provenance detection when PDF-derived grayscale images are subjected to realistic distortions?

To answer this question, the project proceeds through five major tasks before the final report:

1. Repository setup and baseline reproduction
2. Dataset augmentation and complexity simulation
3. Robustness testing under controlled augmentations
4. Additional classifier implementation
5. Comprehensive comparative analysis

This report presents the complete methodology, results, and interpretation of those tasks.

2. Background and Motivation

The motivation for this project lies at the intersection of machine learning and digital forensics. In traditional image classification tasks, models often rely on semantic content such as edges, textures, and object shapes. In this project, however, the objective is different: the grayscale images represent binary encodings, not photographs or natural scenes. That means the “visual”

patterns being learned are really artifacts of file structure, encoding behavior, object placement, and generation pipeline.

This is a valuable forensic problem because provenance detection can potentially assist with:

- attribution of document creation workflows,
- detection of manipulated or regenerated documents,
- identification of suspicious document origins,
- and broader research into machine-readable file signatures.

The project is also an instructive machine learning case study because it highlights how extremely strong clean-data performance can still conceal vulnerability to domain shift. A model that nearly perfectly separates Word and Google Docs on original samples may fail badly if a distortion destroys the specific patterns it learned. Robustness analysis is therefore not a secondary concern here; it is central to understanding whether the system is truly useful.

Another motivation for the project is model comparison. Different classifiers encode different assumptions:

- SVM can learn complex non-linear boundaries,
- SGD is a linear classifier with strong efficiency and surprising effectiveness,
- Random Forest captures non-linear interactions through ensembles of trees,
- and KNN depends directly on distance relationships in feature space.

Because the data are high-dimensional and structurally unusual, comparing these models reveals how classifier choice interacts with representation design.

3. Dataset and Preprocessing

3.1 Original Dataset

The original dataset consisted of grayscale images derived from PDF files created by different applications. For the primary experiments in this report, the two-class setup used:

- **398 Word-generated images**
- **396 Google Docs-generated images**

This produced a total of **794 original images** for the main classification experiments. In earlier setup and augmentation stages, the broader image collection also included Python-generated PDFs, which were preserved in the repository and augmentation outputs. However, the strongest comparative pipeline for Tasks 3–5 focused on the Word-versus-Google classification setting because that was the configuration supported by the baseline and expanded analysis scripts.

Each image was resized to 200×200 grayscale pixels and flattened into a **40,000-dimensional feature vector**. This representation is simple but effective: it standardizes input dimensionality while preserving the broad structural patterns embedded in the binary-to-image conversion.

3.2 Training and Testing Split

For all robustness and comparative analyses, the original dataset was split into:

- **80% training**
- **20% testing**

The split was stratified to preserve class balance. This resulted in:

- **635 training samples**
- **159 test samples**

Most importantly, the classifiers were trained **only on original images**. The augmented datasets were used strictly for evaluation. This is critical because the assignment defines the robustness experiment as a domain-shift problem rather than a data-augmentation-for-training problem.

3.3 Feature Scaling

Feature scaling was applied where appropriate:

- **SVM, SGD, and KNN** used standardized features.
- **Random Forest** used unscaled original features, since tree-based methods are generally insensitive to monotonic feature scaling.

This ensured fairer comparisons among classifiers while respecting each model's strengths and typical preprocessing requirements.

4. Augmentation Methodology

4.1 Required Augmentations

The assignment required exactly five independent augmentations per original image:

1. Gaussian Noise
2. JPEG Compression
3. DPI Downsampling
4. Random Cropping
5. Bit-Depth Reduction.

Each augmentation was applied **independently to the original image**, not chained with other augmentations. This design matters because chained augmentation would make it difficult to attribute degradation to a specific distortion type.

4.2 Parameter Ranges

The implemented augmentations followed the assignment's required ranges:

- **Gaussian Noise:** additive noise with sigma sampled from **[5, 20]**
- **JPEG Compression:** re-encoding with quality sampled from **20–80**
- **DPI Downsampling:** simulated reduction from **300 DPI** to **150 or 72 DPI**
- **Random Cropping:** removal of **1–3%** from each border
- **Bit-Depth Reduction:** reduction from **8-bit grayscale** to **4-bit grayscale**.

4.3 Dataset Expansion

The augmentation process produced:

- **894 original images**
- **4470 augmented images**
- **5364 total images including originals**

This exactly matches the assignment requirement that the final dataset be six times larger than the original, since each original yields five augmented variants in addition to itself.

4.4 Metadata Tracking

A metadata file was generated to document augmentation parameters:

- `augmented_images/augmentation_metadata.csv`

This file records the augmentation type and parameters used for each generated image, making the process reproducible and auditable.

5. Experimental Design

5.1 Task 3 Robustness Design

The robustness experiment followed the required structure:

- Train on the original dataset only
- Test on the original test set
- Test separately on each augmentation type.

This creates a clean framework for measuring how much accuracy is lost when the distribution shifts away from the training condition.

5.2 Task 4 Additional Classifiers

Two additional classifiers from different categories were added:

- **Random Forest** (ensemble tree-based method)
- **KNN** (instance-based method).

Random Forest was selected because it can model non-linear interactions and tends to be robust on structured tabular data. KNN was selected because it provides a useful contrast: instead of learning explicit decision boundaries, it classifies based on distances between examples.

5.3 Task 5 Comprehensive Analysis

The final comprehensive analysis compared:

- **SVM**
- **SGD**
- **Random Forest**
- **KNN**

Each model was evaluated across:

- original data,
- Gaussian noise,
- JPEG compression,
- DPI downsampling,
- random crop,
- bit-depth reduction.

For each condition, the following were computed:

- accuracy,
- precision,
- recall,
- F1-score,
- confusion matrices,
- confidence intervals,
- and pairwise bootstrap significance tests.

6. Results

6.1 Baseline and Original-Set Performance

The clean-data results on the original test set were:

Model	Accuracy
SGD	0.9937
Random Forest	0.9874
SVM	0.9686
KNN	0.4969

These results show a clear separation between the strongest and weakest models. SGD achieved the best performance, slightly outperforming Random Forest. SVM also performed strongly, though not at the same level as SGD. KNN failed badly, performing at approximately random chance.

6.2 Task 3 Robustness Results

For the two baseline models, Task 3 produced the following main findings:

SVM

- Original: **0.9686**
- Gaussian noise: **0.9686**
- JPEG compression: **0.9686**
- DPI downsampling: **0.4969**
- Random crop: **0.5723**
- Bit-depth reduction: **0.9686**

SGD

- Original: **0.9937**
- Gaussian noise: **0.9937**
- JPEG compression: **0.9937**
- DPI downsampling: **0.9560**
- Random crop: **0.4969**
- Bit-depth reduction: **0.9937**

The largest degradations were:

- **SVM:** DPI downsampling
- **SGD:** random cropping

These findings already suggested that structural distortion is much more dangerous than pixel-level distortion.

6.3 Task 4 Additional Classifier Results

The two added classifiers produced the following original-set performance:

Random Forest

- Best parameters:
 - `n_estimators = 100`
 - `max_depth = None`
 - `min_samples_split = 2`
- Accuracy: **0.9874**

KNN

- Best parameters:
 - `metric = euclidean`
 - `n_neighbors = 3`
 - `weights = uniform`
- Accuracy: **0.4969**

Random Forest performed exceptionally well, rivaling SGD. KNN failed almost completely, predicting one class for nearly all examples.

6.4 Task 5 Comprehensive Results

The full Task 5 analysis confirmed the following trends:

SVM

- Strong on original, noise, compression, and bit-depth reduction
- Severe degradation under downsampling
- Major degradation under cropping

SGD

- Best original performance overall
- Extremely robust to noise, compression, and bit-depth reduction
- Slight degradation under downsampling
- Severe degradation under cropping

Random Forest

- Nearly as strong as SGD on original data
- Very robust to noise and compression
- Moderate degradation under downsampling
- Severe degradation under cropping
- Small degradation under bit-depth reduction

KNN

- Near-random performance across all conditions
 - No meaningful robustness because baseline performance was already poor
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7. Robustness Analysis

7.1 Why Some Augmentations Had Little Effect

Three augmentations had very limited impact on the top-performing models:

- Gaussian noise
- JPEG compression
- Bit-depth reduction

These augmentations alter local pixel intensities, but they do **not fundamentally change global structure**. In this project, provenance appears to be encoded more in large-scale layout and binary-image pattern organization than in precise grayscale values. As a result, the dominant signal survives noise injection, moderate JPEG recompression, and tonal quantization.

This is an encouraging result because it implies that the system is robust to realistic low-level distortions that might occur during storage, transmission, or ordinary preprocessing.

7.2 Why Downsampling Hurt SVM So Much

DPI downsampling caused the largest degradation for SVM. This can be explained by the fact that downsampling removes high-frequency detail and smooths structure. If the SVM relies on subtle structural patterns or fine-grained binary-image textures, then aggressive reduction from 300 DPI to 150 or 72 DPI destroys that information. The confusion matrix under this condition showed that SVM collapsed toward predicting a single class, indicating that the learned boundary no longer separated the classes meaningfully.

Interestingly, SGD was much more resilient to downsampling. This suggests that the linear classifier may be relying on broader, more distributed structure that survives resolution loss better than the more specific boundary learned by SVM.

7.3 Why Cropping Was So Damaging

Random cropping was the most harmful augmentation overall, especially for SGD and Random Forest. Cropping removes border regions and modifies alignment, margins, and the global placement of binary-image patterns. Because provenance cues may include positional or layout-specific signatures, cropping destroys exactly the kind of information that these models appear to use.

This is a particularly important result from a forensic perspective. It suggests that provenance classifiers may be vulnerable if a document image is partially captured, trimmed, or missing consistent border structure.

7.4 Why Bit-Depth Reduction Was Mostly Harmless

Bit-depth reduction had almost no effect on SVM and SGD, and only a small effect on Random Forest. This implies that the key features are not encoded in delicate grayscale transitions. Instead, they are likely represented through more stable structural distributions that remain visible even after reducing tonal resolution from 8-bit to 4-bit grayscale.

8. Statistical Analysis

The comprehensive analysis used bootstrap-based significance testing for pairwise model comparisons on the original test set. This choice is appropriate because it provides both confidence intervals and hypothesis testing without requiring overly strong parametric assumptions.

8.1 Pairwise Results

Key comparisons:

- **SVM vs SGD:** $p = 0.022$
- **SVM vs Random Forest:** $p = 0.104$
- **SGD vs Random Forest:** $p = 0.774$
- **All comparisons vs KNN:** $p \approx 0.000$

8.2 Interpretation

These results indicate:

- SGD significantly outperformed SVM.
- Random Forest did not differ significantly from SGD.
- KNN was significantly worse than all strong models.

This statistical layer strengthens the empirical findings by showing that the ranking is not just based on small accidental differences in accuracy. It also supports the claim that both SGD and Random Forest are top-tier choices for this problem.

9. Discussion

This project produces several important conclusions about machine learning for PDF provenance detection.

First, the overall approach is clearly valid. Multiple models achieved very high accuracy on the original test set, confirming that the grayscale binary-image representation contains strong provenance signals. This alone is a meaningful result because it shows that file-generation differences are detectable in a machine-learnable way.

Second, robustness is highly dependent on the type of distortion. The system is not uniformly fragile; instead, it is selectively vulnerable. Pixel-level degradations such as noise and compression matter much less than structural transformations such as cropping and downsampling. This is a crucial insight because it tells us what kinds of real-world distortions are likely to be most dangerous in forensic workflows.

Third, classifier choice matters substantially. SGD and Random Forest emerged as the strongest options. SVM was still effective, but less robust under certain structural distortions. KNN was not appropriate for this problem at all. Its failure is consistent with the **curse of dimensionality**: in a 40,000-feature space, distance measures become much less meaningful, which makes nearest-neighbor reasoning unreliable.

Fourth, the project highlights the difference between accuracy and forensic usefulness. A classifier can achieve near-perfect performance on clean samples while still breaking under plausible distortions. This reinforces why robustness testing is essential before making strong practical claims.

There are also limitations. The current experiments rely on a relatively simple resizing-and-flattening representation rather than more advanced feature extraction or deep learning. The analysis also focused heavily on the Word-versus-Google setting for robustness and comparative evaluation, even though Python-generated documents were part of the broader dataset. Future work could extend the full robustness pipeline to three-class classification and compare whether programmatically generated PDFs show different vulnerability patterns.

10. Conclusion

This project successfully scaled the ForensicsDetective framework through augmentation, robustness analysis, and broader model comparison. The main findings are:

1. **PDF provenance classification is highly accurate on clean data.**
2. **SGD and Random Forest are the strongest overall models.**
3. **Gaussian noise, JPEG compression, and bit-depth reduction have little effect on the best models.**
4. **Random cropping and DPI downsampling are the most damaging augmentations.**
5. **KNN is ineffective in this high-dimensional setting.**

6. Statistical testing confirms that SGD significantly outperforms SVM, while SGD and Random Forest are statistically similar.

Overall, the study demonstrates that provenance classification from PDF-derived grayscale images is both powerful and sensitive to structural distortion. The strongest classifiers capture document-generation signatures effectively, but practical forensic deployment would need to account for the sharp performance degradation caused by cropping and downsampling. These results provide a strong foundation for future work in document forensics, especially work that aims to improve robustness under realistic acquisition conditions.

References

Course materials and assignment specification for EAS 510.

Scikit-learn documentation for SVM, SGD, Random Forest, KNN, and evaluation metrics.

OpenCV and Pillow documentation for image processing and augmentation workflows.